BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA

Order Instituting Rulemaking to Modernize the Electric Grid for a High Distributed Energy Resources Future.	Rulemaking 21-06-017 (Filed June 24, 2021)

COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE PROPOSED DECISION ADOPTING IMPROVEMENTS TO DISTRIBUTION PLANNING AND PROJECT EXECUTION PROCESS, DISTRIBUTION RESOURCE PLANNING DATA PORTALS, AND INTEGRATION CAPACITY ANALYSIS MAPS

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In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC") hereby submits these comments on the *Proposed Decision Adopting Improvements to Distribution Planning and Project Execution Process, Distribution Resource Planning Data Portals, and Integration Capacity Analysis Maps* ("PD") issued by Commissioner Houck on September 13, 2024.

I. INTRODUCTION.

VGIC commends the Commission for beginning the much-needed process of updating California's distribution planning paradigms and processes to accommodate the high amounts of additional load and generation connecting to California's distribution grid. While many sectors are electrifying, transportation electrification, especially for medium- and heavy-duty vehicle sites, is rapidly becoming one of the largest sources of new load with projects that often exceed several megawatts ("MW") and can be deployed with relatively short customer-side construction timelines. Planning for and managing electric vehicle ("EV") load is critical. California is currently experiencing an energy affordability crisis, and while much of the increase in electric utility rates to date has been driven by wildfire mitigation costs, distribution system upgrades will add significant new costs. At the same time, new load and increased ***off-peak*** kWh sales may put downward pressure on rates. Ensuring that kWh sales are maximized while distribution upgrades are minimized to the extent possible is key to ensuring affordability. Notably, vehicle-grid integration ("VGI") initiatives that encourage more efficient use of electrical infrastructure, like managed charging programs and bidirectional charging efforts, are one of the few tools in the toolkit to unlock downward pressure on rates.

The PD makes many essential modifications to improve distribution optimization in the long term. Generally, VGIC supports the PD's distribution planning proposals, including better incorporating known load data into distribution plans, utilizing more recent Integrated Energy Policy Report ("IEPR") load forecasts, extending the distribution planning horizon, and other modifications to make distribution planning more proactive. VGIC also supports many of the proposed modifications to the Integrated Capacity Analysis ("ICA") maps and the focus of the Commission on increasing accuracy for the Load ICA in particular. VGIC strongly supports the Commission adopting our recommendation to establish a venue for regular discussion of the ICA maps in the form of quarterly workshops.

However, the PD does not adequately consider the actions the utilities can take *now* to allow additional EV load to come online in the near term and to further optimize the integration of EV charging into the existing distribution system. In particular, flexible service connections and Limited Load Profiles ("LLPs") have the potential to enable additional load to come online in the near term and provide data and experiences that can be used to optimize long-term VGI. To that end, VGIC offers the following comments on the PD:

- The Commission should establish standardized flexible service connection offerings.
- The PD should enable the implementation of Limited Load Profiles in the ICA maps.

- The Electrification Impact Study ("EIS") Part 2 Study should include multiple managed charging scenarios and considerations for EV load mitigation strategies.

II. THE COMMISSION SHOULD ESTABLISH STANDARDIZED FLEXIBLE SERVICE CONNECTION OFFERINGS.

The Commission has recently focused on energization timelines and ensuring that new loads can connect to the grid in a timely manner. The Commission has taken action to implement average and maximum energization timelines for the investor-owned utilities ("IOUs"), but timelines for constructing grid upgrades are still multiple years long.[1] VGIC strongly believes that the Commission should take action to address near-term needs by enabling the widespread use of flexible service connections. These connections allow customers to energize their load in a flexible manner, typically to accommodate distribution grid constraints.

The Staff Proposal discusses flexible service connections as bridging strategies, stating that projects triggering distribution upgrades can be energized "temporarily, until the upgrade is completed, using temporary load flexibility via voluntary load curtailment for specific limited times or DER assets such as mobile batteries."[2] As discussed in the PD, PG&E and SCE are considering different flexible service connections and conducting pilots for different types of bridging solutions. These plans should be further outlined for stakeholders, and VGIC supports the PD's requirement that the utilities submit their plans for bridging strategies by December 15, 2024. The Staff Proposal also recommends that the utilities provide updates on implementing their plans in the annual DDOR. The PD adopts the requirement for utilities to provide a bridging strategies proposal, but declines to require their implementation or annual reporting. The PD states

[1] D.24-09-020 at 47.
[2] Staff Proposal at 54.

that this is because "processes for long-term, scalable load management as a bridging solution do not exist for evaluation and execution."[3] However, the Commission could begin the regulatory process to gather proposals for evaluation.

The Commission should take broader action now to create widely available flexible service connections in order to enable near-term and longer-term ratepayer savings. Flexible service connections leverage technologies that are available today, and allow load to come online and increase efficient use of existing electrical infrastructure. Critically, electrification projects leveraging flexible service connections, by definition, add off-peak load, since service upgrades need to be made to accommodate peak load. **This makes flexible service connections one of the only tools the Commission has to curb the outsized ratepayer impacts already felt by millions of Californians.** While important bridging solutions, the development of flexible service connections will lead to the creation of different forms of Limited Load Profiles ("LLPs"), which can be used to establish long-term agreements for flexible load.

Enphase has raised this issue in this instant proceeding and the Energization Timelines proceeding (R.24-01-018), submitting motions to open a regulatory process to standardize flexible service connections and implement LLPs.[4] VGIC strongly believes that Enphase's motion should be approved in the Energization Rulemaking. Importantly, the Commission must not further postpone addressing these issues by debating the proper regulatory venue for discussion. As the Commission looks to expand the development of flexible service connections, the plans submitted

[3] PD at 89.

[4] *Motion of Enphase Energy, Inc. to Amend Scoping Ruling and Schedule to Address Optional Flexible Connection Agreements and the Underwriters' Laboratory 3141 Standard for Power Control Systems in Investor-owned Utilities' Energization Rules* filed by Enphase Energy in R.24-01-018 on September 11, 2024.

by the IOUs on bridging strategies can be used as the starting point for broader party proposals, workshops, and other discussions.

III. THE PD SHOULD ENABLE THE IMPLEMENTATION OF LIMITED LOAD PROFILES IN THE ICA MAPS.

VGIC encourages the Commission to direct the utilities to begin planning for incorporating Limited Load Profiles in the Load ICA maps now. As discussed above, it is important for the Commission to leverage widespread flexible service connections as soon as possible. It is likely that this will take the form of LLPs with import limits that vary throughout the day, similar to the Limited Generation Profiles ("LGPs") with varying export schedules. The PD adopts Staff's recommendation to modify the ICA maps to enable straightforward customer creation of LGPs, and VGIC recommends that the Commission also adopt plans to enable similar straightforward integration for Limited *Load* Profiles.

Once a broader flexible service connection framework is adopted, customers will need to have access to information on grid constraints throughout the day and year in order to decide whether an LLP will work for them. Having data be publicly accessible in the Load ICA will be the easiest way for customers to conduct their own inquiries into how different sites can utilize LLPs. The ICA maps can also be valuable tools for utilities to use internally to gather information to present to customers. VGIC supports the PD's requirement for PG&E to utilize their Load ICA to reduce the length of their intake process for energization applications. Broadly, VGIC sees a future where both the utilities and customers utilize the ICA maps to aid in the implementation of LLPs for a given site.

Given the amount of work and data that will be required to integrate LLPs into the ICA maps, VGIC suggests that the Commission signal its intention to enable this use case in this decision. This way, the utilities can provide updates on how they plan to implement LLPs in their Annual Load ICA Refinements Reports. There may also be opportunities to leverage the work being done for other Load ICA refinements or general ICA refinements to gather the necessary information for LLPs, reducing the need for the IOUs to go back to collect this data later.

IV. THE ELECTRIFICATION IMPACT STUDY PART 2 STUDY SHOULD INCLUDE MULTIPLE MANAGED CHARGING SCENARIOS AND CONSIDERATIONS FOR EV LOAD MITIGATION STRATEGIES.

VGIC supports the inclusion of the Load Flexibility Assessment in Part 2 of the Electrification Impact Study ("EIS") being conducted by the utilities. The PD states that the Part 2 EIS will "estimate and assess potential impacts (e.g., potential costs of upgrading the primary and secondary distribution grid) of meeting electrification needs under multiple scenarios."[5] VGIC supports the inclusion of multiple scenarios in the Part 2 study, and we reiterate our previous recommendations to include the following managed charging scenarios:

- Baseline EV Load Management, based on the assessment that has already been completed in Part 1 of the study.
- Mid EV Load Management, based on updated IEPR assumptions compared to the Part 1 study, which reflect higher TOU rate participation from EVs.
- High EV Load Management, based on the inputs and assumptions for high EV load management used in the June 2023 Updated Integrated Resource Planning Inputs and Assumptions document.[6]

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[5] PD at 95.

[6] Inputs & Assumptions, 2022-2023 Integrated Resource Planning (IRP), June 2023. https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/energy-division/documents/integrated-resourceplan-and-long-term-procurement-plan-irp-ltpp/2023-irp-cycle-events-andmaterials/draft_2023_i_and_a.pdf

- High EV Load Management + EV Charger Export, based on the inputs and assumptions used in the June 2023 Updated Integrated Resource Planning Inputs and Assumptions document.[7]

It is clear that EVs will be adding a significant portion of the new load that is triggering distribution upgrade needs, and it is therefore clear that management of EV charging load is key to reducing the overbuild of the distribution system and reducing the ratepayer costs of this build out. For example, the EIS Part 1 Study and Cal Advocates' Distribution Grid Electrification Model ("DGEM") Study have drastically different cost estimates to upgrade the distribution system because their assumptions around EV charging profiles differ, leading to varying load peaks and distribution needs. Including multiple scenarios will allow the Commission and stakeholders to evaluate how managing charging profiles can help reduce ratepayer costs. This will influence California's efforts to expand managed charging programs and provide a more thorough assessment of the ratepayer costs and benefits for different managed charging initiatives.

For this reason, VGIC also continues to recommend that the Part 2 Study evaluate the potential for software- and hardware-based solutions to manage EV charging load specifically for local grid constraints (i.e., as opposed to system-level needs), including:

- Platforms to manage EV charging load on the distribution system.
- Site-level EV load management utilizing power-sharing software across several chargers at a single customer site.
- Storage-backed EV charging and DER-paired EV charging.

[7] Ibid.

This will provide California with valuable information about how to shape load and maximize California's ability to transition to EVs while also minimizing costs to ratepayers as much as possible.

V. CONCLUSION.

VGIC appreciates the opportunity to provide these comments on the Proposed Decision. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

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Date: October 3, 2024